



Thermal analysis and optimization of an EV battery pack for real applications

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ABSTRACT

Battery packs are extensively used in electrical vehicles (EV) to avoid environmental pollution. The safety, aging and life of battery pack are significantly related to its thermal behavior. This work concerns with thermal analysis and optimization of an EV battery pack for real engineering applications. The Bernardi's heat generation model with the consideration of reversible heat is used and validated by tests and numerical simulation on battery cells. Then the validated cell model is applied to an EV battery pack with cooling system underneath for the study of thermal behavior at two extreme operation conditions, and numerical results are in good agreement with the test results. Several key factors important to the thermal behavior of battery packs are studied by performing sensitivity analysis at these two operation conditions, and it is found that the inlet temperature and the inlet flow are the two most significant factors. For better thermal performance of the EV battery pack, optimization analysis at two extreme operation conditions is conducted to determine the optimal parameters for the inlet temperature and the inlet flow, and this optimization process provides a means to find out the best input parameters for real engineering problems.

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1. Introduction

Battery pack is one of the most important components of electrical vehicles (EVs), and the performance and cost of EVs is significantly affected by the performance and life of battery packs. The battery pack, however, is sensitive to operating temperature on its performance [1,2]. Specifically, when batteries are working in the condition of high or low temperature or battery cells are present with unbalanced temperature, it will lead to accelerated battery aging, decreased discharging performance and even worse, serious battery safety accidents, e.g., thermal runaway and propagation [3,4]. Therefore, thermal management analysis is crucial for studying thermal behavior of battery packs.

In thermal analysis of battery packs, the generated heat from battery cells is an important source of heat. So far, there are two means to obtain this physical parameter. The first one is based on thermal-electro-chemical battery model, which enables our attention to be paid on the underlying mechanism of heat generation; however, this model requires a large number of electro-chemical parameters and these unknowns are difficult to determine. The second one is the simplified Bernardi's heat generation model [5] which is widely used because of physical-based prop-

erty and few parameters are required. In the Bernardi's heat generation model, the entropy coefficient is an important parameter as it is related to reversible heat. This important parameter, however, is difficult to test and determine and sometimes neglected or assumed to be a constant. Apparently, these simplified treatments will lead to the bias of the generated heat between the calculation and the test, and thus it is necessary to measure and calculate the real values of the entropy coefficient. To obtain this parameter, small-capacity cells of 18650 were selected by Al Hallaj et al. [6], and medium-size capacity cells were selected by Zhang et al. [7], but unfortunately, large-capacity cells are rarely used to conduct research as from literature. Considering a large-capacity cell has less heat releasing rate due to the small ratio of surface to volume, we use a large-capacity cell of 153 Ah to investigate the entropy coefficient for thermal management analysis of a battery pack in this research.

A great number of researches have been focused on the thermal behavior of a single cell [6,8,9]; however, single cell neither reflects the temperature-rising property of module and pack, nor takes account for the heat dissipation effect of cooling system which is crucial to the battery pack. Smith et al. [10] hence put emphasis on a battery module with the capacity of 25 Ah and studied the influence of inlet temperature and flow flux to the battery temperature; however, the temperature difference between battery modules, which is important to battery pack, cannot be obtained. Chen

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et al. [11] made a preliminary study at a battery pack level with focus on the effect of radiation heat transfer to battery pack.

In this research, a real EV battery pack with cooling system underneath from the GAC Motor is employed as our model for thermal analysis and optimization at two extreme operation conditions. Firstly, the accurate heat generation model of a single cell is formulated and validated by tests with the consideration of reversible heat. Specifically, the internal resistance and open circuit voltage (OCV) of cells are acquired under the condition of varying temperature and state of charge (SOC) by tests. To obtain the heat generation of cells, the irreversible heat, which is based on the internal resistance, and the reversible heat, which is related to the entropy coefficient, are calculated by means of the Bernardi's model [5]. It should be noted that the entropy coefficient is obtained based on the test data from a Ni-Co-Mn lithium-ion battery with the capacity up to as large as 153 Ah. The heat generation and temperature of the battery cell are tested by the extended volume-accelerating rate calorimetry (EV-ARC) at different discharging rates, and it would find that the tested heat generation agrees well with that obtained by the Bernardi's model and the tested temperature is in good agreement with the simulation results.

Then, on the basis of the validated cell models, a real EV battery pack with cooling system underneath is established for 3D thermal analysis and optimization. Numerical results from two realistic operation conditions are in good agreement with the tests. Six factors, i.e., inlet temperature and flow of coolant, conductive coefficient of cooling pad, conductive coefficient in cell's X, Y and Z-directions, are compared for the impact on the maximum temperature (T_{max}), the maximum heating time (t_{max}) and the maximum temperature difference (ΔT_{max}). Regarding to the application of optimization to battery packs, a number of works have been focused on either structural optimization of cooling plates [1,12,13] or battery cell spacing [14] or the widths of plenum in an air-cooled battery packs [15]. Those works are beneficial to the design of battery packs in an early stage; however, those optimal designs sometimes have to be compromised due to the layout of other structural components or concerns of electromagnetic compatibility. To the optimization of an integrated EV battery pack for real applications, we mainly consider to find out the best input parameters of inlet flow and inlet temperature for the optimal thermal behavior of the EV battery pack.

The remainder of the paper is organized as follows. Firstly, the governing equations of energy conservation for the battery pack and coolant are given in Section 2. Then the Bernardi's heat generation model is formulated and the model is validated by tests and numerical simulation at different discharging rates in Section 3. In Section 4, the battery cell model which takes the heat generation model into account is applied to the thermal analysis of a real EV battery pack at two operation conditions. Furthermore, a number of input parameters are investigated to study their sensitivity to the thermal behavior of the battery pack model, and optimization analysis is performed to find the optimal input parameters. At last, some conclusions are drawn.

2. Governing equations

The temperature distribution is of importance to the thermal analysis of EV battery packs. The governing equation of energy conservation for the heat transfer problems of battery packs can be written as

$$\rho c_p \frac{\partial T(x, t)}{\partial t} = \frac{\partial}{\partial x_i} \left(k_{ij} \frac{\partial T(x, t)}{\partial x_j} \right) + q_v \quad (1)$$

in which, T is the temperature and it varies with spatial position vector $\mathbf{x}(x, y, z)$ and time t ; ρ and c_p are density and specific heat

capacity of the battery, respectively; x_i indicates the i th component of $\mathbf{x}$; k_{ij} is heat conductivity and the battery material is usually orthotropic and hence $k_{ij} = 0$ for $i \neq j$; q_v is the volumetric heat source in the cell, which is generated by electrochemical enthalpy change and internal thermal resistance; the subscripts of i, j are the index and $i, j = 1, 2, 3$ for three dimensional problems.

The initial condition of the temperature for battery pack is given as

$$T(x, y, z, 0) = T_0 \quad (2)$$

where T_0 is the initial temperature of the battery.

The boundary condition related to thermal analysis of battery pack and cells includes convection, adiabatic and radiation boundary conditions. The heat transfer via convection occurs between battery pack and environment or coolant. The convection boundary condition is expressed as

$$-k_{ij} \frac{\partial T}{\partial x_j} n_i = h(T - T_\infty) \quad (3)$$

in which, h indicates heat exchanging coefficient between the battery pack and environment or coolant; n_i is the component of the outward surface normal vector $\mathbf{n}$ and T_∞ indicates the ambient temperature or the coolant temperature. The adiabatic boundary condition takes a simplified form of the convection boundary condition and is expressed as

$$-k_{ij} \frac{\partial T}{\partial x_j} n_i = 0 \quad (4)$$

which means no heat exchange occurs on certain boundary.

The radiation boundary condition is expressed as

$$-k_{ij} \frac{\partial T}{\partial x_j} n_i = \varepsilon \sigma (T^4 - T_\infty^4) \quad (5)$$

where ε is the emissivity of radiant surface and σ the Stefan-Boltzmann constant.

The EV battery pack is normally working with the liquid cooling system underneath, which is used to cool or heat cells in different driving conditions. Liquid coolant is usually a mixture of water and glycol. The energy conservation equation of liquid coolant can be written as

$$\frac{\partial (\rho_c c_c T_c)}{\partial t} + \rho_c c_c T_c \frac{\partial v_i}{\partial x_i} = \frac{\partial}{\partial x_i} \left(k_c \frac{\partial T_c}{\partial x_i} \right) \quad (6)$$

where ρ_c , c_c and k_c are the density, specific heat capacity and thermal conductivity of the isotropic coolant, respectively; T_c is the temperature of coolant and is related to spatial and time; v_i the i th component of the velocity vector $\mathbf{v}$ of coolant.

3. Heat generation model of battery cells

The Bernardi's heat generation model is formulated at first, and then the EV-ARC test results and numerical results are employed to validate the heat generation model.

3.1. Formulation of the heat generation model

To analyze the thermal behaviour of the battery pack, the heat generation model of battery cells is critical. Generally, there are two categories of heat generation models. The first one is based on thermo-electrochemical battery model [16] and studies the mechanism of heat generation. However, this model requires a large number of electrochemical parameters of battery, and moreover, these parameters are hard to determine. The second one is the widely-used Bernardi's heat generation model [5] and only few parameters are required. For simplicity, the latter one is employed in this research.

According to the Bernadi's heat generation model [5], its simplified form can be given as

$$Q = I \left((U_{oc} - V) + T \frac{dU_{oc}}{dT} \right) \quad (7)$$

where Q is the total heat generation rate from the battery cells; I is the working current of the battery; U_{oc} is the open voltage of the battery circuit; V is the working voltage of the battery; T is the temperature of the battery and dU_{oc}/dT means entropy coefficient which in general describes the entropy change of the battery cells [17].

As considering Ohm's law,

$$U_{oc} - V = IR \quad (8)$$

in which, R represents the equivalent internal resistance of the circuit, and is usually calibrated by the current-off method [18]. Rearranging Eqs. (7) and (8) yields

$$Q = I^2 R + IT \frac{dU_{oc}}{dT} \quad (9)$$

It should be noted that I is positive when charging and negative when discharging.

The first term on the right hand side of Eq. (9) is Joule heating. It is in fact the irreversible heat generation rate Q_{ir} and caused by battery internal resistance. The second term on the right hand side of Eq. (9) implies reversible heat generation rate Q_r , which represents the charging and discharging characteristics of cells. With the decrease of the discharging rate, the working current I decreases. As the irreversible heat is the square of I (from the Eq. (9)), the change of the irreversible heat generation rate Q_{ir} is much greater than the change of the reversible heat generation rate Q_r which is linear to I . Therefore, it can be found that Q_{ir} is significant to large discharging rate and on the contrary, Q_r is significant to small discharging rate. This conclusion will be used later in Section 3.3.

To determine the reversible heat generation rate Q_r , the entropy coefficient dU_{oc}/dT has to be known; however, this unknown variable cannot be directly obtained from the test. The potentiometric method and the calorimetric method are often applied for its measurement [7]. As concluded by Thomas et al. [19] that the potentiometric method is more accurate than the calorimetric method, the potentiometric method is employed here. Two steps need to be carried out to obtain the entropy coefficient. Firstly, the relationship between the OCV and the SOC of battery cells at different temperature is obtained by the offline OCV tests [20]. Secondly, the OCV measurement data at each SOC are fit and normally found to be approximately linear to the temperature, thus the slope, i.e., the dU_{oc}/dT , can be derived. A correlation of the entropy coefficient dU_{oc}/dT to the SOC is then obtained by the test and plotted as shown in Fig. 1. Note that this curve of entropy coefficient will be applied to the heat generation model of battery cells in Section 3.2.

The OCV measurements require long relaxation time [21], which requires from a few minutes to even hours, and the accuracy depends on hysteresis phenomenon and recovery effect [17,22,23];

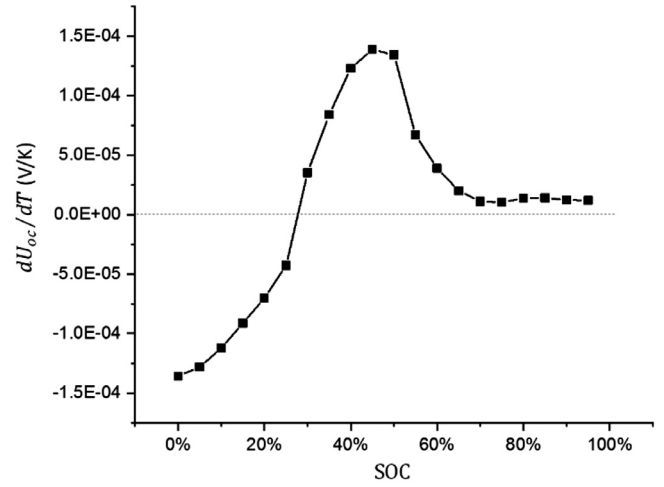


Fig. 1. Correlation of entropy coefficient to SOC.

therefore, the entropy coefficient is hard to obtain and usually neglected or assumed to be a constant instead, leading to a zero or inaccurate reversible heat generation rate Q_r . Details of the impact from these simplified treatments are discussed in Section 3.2.2.

3.2. EV-ARC tests of battery's heat generation

The EV-ARC is employed to test the specific heat capacity (C_p) and the heat generation (W) of battery cells, where the nominal capacity (C) of cells is 153 Ah. To test C_p , a battery sandwich structure that a heating pad is sandwiched by two cells as shown in Fig. 2(a) is placed in the container of an EV-ARC. The heating pad is wired to an external DC power supply with a constant heating power P . The EV-ARC is working in an isothermal-fixed mode that ensures the battery sandwich is always in an adiabatic condition. To record the temperature of the cell, a thermocouple of the negative temperature coefficient (NTC) sensor is placed on the middle of the cell's large face. Then the experimental values of battery's temperature can be obtained as a function of time. Therefore, according to the energy conservation that the total external heat input to the battery sandwich is transformed to the temperature rise (ΔT) of the structure itself, the following equation satisfies

$$P\Delta t = C_p m_s \Delta T \quad (10)$$

where Δt is the heating time and m_s the mass of the sandwich structure. Rearranging Eq. (10), yields

$$C_p = \frac{P}{m_s} \cdot \frac{1}{\Delta T / \Delta t} \quad (11)$$

As the temperature rise ΔT and the heating time Δt can be directly obtained from the EV-ARC test data, the specific heat capacity C_p can then be calculated by Eq. (11).

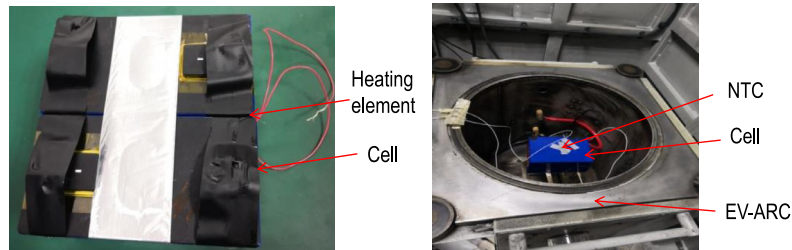


Fig. 2. Experimental set-up for testing cells' (a) specific heat capacity and (b) heat generation by the EV-ARC.

To test the heat generation W of the battery cell, a cell is placed in the container of an EV-ARC and wired to an external device as shown in Fig. 2(b). To test the heat generation behavior of the battery cell, the cell is discharging with different rates of 1.2 C, 0.67 C and 0.2 C. Because of discharging, the battery cell would generate heat and so that the temperature arises. A thermocouple sticks to the middle of the cell's large face to record the temperature variation, and to obtain effective temperature data, the EV-ARC is again working in an isothermal-fixed condition. As the specific heat capacity C_p is obtained from the above-mentioned experiment, the heat generation W can be obtained as a relation of the temperature rise ΔT as follows

$$W = C_p m_c \Delta T \quad (12)$$

where m_c denotes the mass of the cell. It should be mentioned that the heat generation W of the tested cell is contributed by both the irreversible heat generation rate Q_{ir} and the reversible heat generation rate Q_r .

3.3. Theoretical validation of the heat generation model

The results of heat generation against the depth of discharge (DOD) at different discharging rates (1.2 C, 0.67 C and 0.2 C) are obtained by the EV-ARC tests and plotted as shown in Fig. 3. For the test results, the heat generation (expressed by Eq. (12)) is obtained by the experiment set-up as shown in Fig. 2(b), and the DOD has a relation with SOC, i.e., $SOC = 1 - DOD$. The SOC is related to its initial state (SOC_{ini}) and the percentage that the battery discharged capacity (C_d) to the nominal capacity (C), i.e., $SOC = SOC_{ini} - C_d/C$, where the battery discharged capacity (C_d) is a product that the battery discharging rate (I) multiplies the discharging time in hours. For the theoretical results, the heat generation rate (Q) is expressed by Eq. (9), where the temperature (T) of cell can be recorded by experiment as shown in Fig. 2(b) and the entropy coefficient dU_{oc}/dT can be obtained from Fig. 1.

The theoretical results from the Bernadi's model with or without considering reversible heat are compared in Fig. 3, from which it can be found that the results of the heat generation agree much better with the test results at three different discharging rates when the reversible heat generation rate Q_r is taken into account. Moreover, the agreement between the test results and the theoretical results without considering Q_r gets worse with the decreasing discharging rate. It should be noted that from the figure, the increase of heat generation results become slower between the 40% and 70% DOD. This is because the entropy coefficient is positive (as shown in Fig. 1) when the SOC ranges from 30% to 60%. In a discharging condition, the working current I is negative and so that the reversible heat Q_r contributes negatively to the total generated heat. In a particular case of discharging rate being 0.2 C, the negative contribution of Q_r is greater than the irreversible heat Q_{ir} , thus Q drops in a certain range where the entropy coefficient is positive, which corresponds to the conclusion as mentioned in Section 3.1 that the Q_r plays a more significant role than the Q_{ir} in the case of small discharging rate.

The errors that defined by the current gap (between test and theoretical results with or without considering reversible heat) to the total heat accumulation during the course are also plotted in Fig. 3, where it is clear that the error for theoretical result with considering reversible heat is much smaller than that from theoretical result without considering reversible heat for all three discharging rates cases. The reason to select the total heat accumulation for error calculation is that the entire period of heat generation is important and used for validation.

From the above discussion, it is summarized here that the heat generation model is accurate and the reversible heat cannot be neglected especially for cases with small discharging rates.

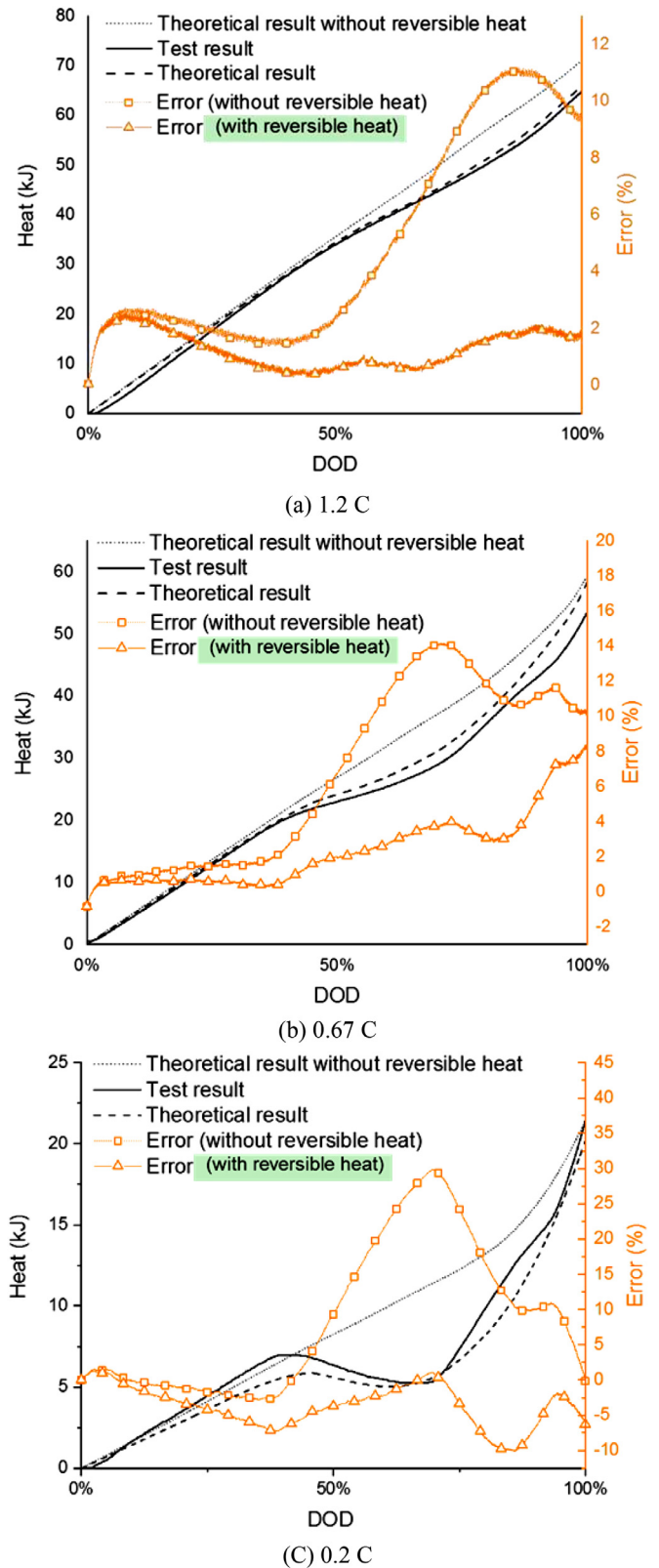


Fig. 3. Correlation of the heat generation to the depth of discharge at different discharging rates.

3.4. Numerical validation of the heat generation model

Battery cells are used for numerical validation and the heat generation model formulated in Section 3.1 is employed with tak-

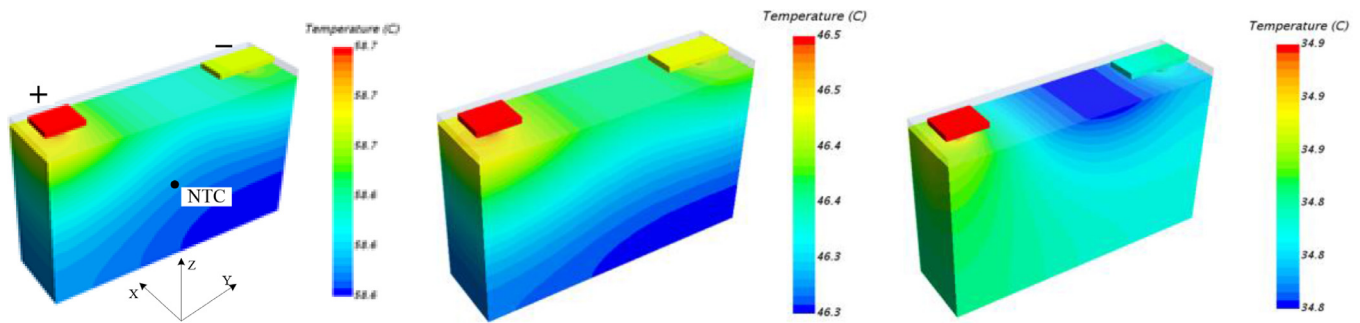


Fig. 4. Temperature distribution of the cell at different discharging rates: (a) 1.2 C, (b) 0.67 C and (c) 0.2 C.

Table 1

Technical parameters for a 153 Ah Lithium-ion battery.

Parameters	Values
Rated capacity	153.0 Ah
Rated voltage	3.7 V
Internal resistance	$\leq 0.8 \text{ m}\Omega$
Specific heat capacity	1000.0 J/kg/K
Density	2277.0 kg/m ³
Size of cell (length $\times$ width $\times$ height)	79 $\times$ 148 $\times$ 95 mm
Thermal conductivity (X direction) k_x	2 W/m/K
Thermal conductivity (Y direction) k_y	20 W/m/K
Thermal conductivity (Z direction) k_z	15 W/m/K

ing account of the reversible heat. Physical and thermal properties of cells are shown in Table 1. Cells are discharging in an adiabatic boundary condition (shown in Eq. (4)) at different discharging rates with the SOC changing from 100% to 0. NTC sensors are placed in the center of cells' large surface to record the temperature variation.

As from the temperature distribution in Fig. 4, the maximum temperature decreases with less discharging rate, which is in agreement with the interpretation of the heat generation model. As for each figure, the maximum temperature appears in the positive electrode because it is made of aluminum and the resistance is larger compared to the other components of cells. By looking into the temperature variation of the center of the large surface, it can be found that the numerical results agree well with the test results by the NTC sensor at three different rates as shown in Fig. 5. The relative errors between the numerical and test results in the

final state (fully discharged) are within 5%. The errors that defined by the current gap (between simulation and test results) to the total heat accumulation during the course are also plotted in Fig. 5, where it is clear that the errors for simulation results at three different discharging rates are small. The reason to select the total heat accumulation for error calculation is that the entire period of heat generation is important and used for validation. As the DOD increases from 40% to 70%, the trend of both the numerical and test results get slower, which is in good agreement with the theoretical validation, and the reason to this change is also the entropy coefficient being positive in this range. This battery cell's model is then to be applied to the battery pack's model in the next section.

4. Thermal analysis and optimization for an EV battery pack

The above-mentioned cells and the heat generation model are applied to a real EV battery pack for 3D thermal analysis and optimization. The 3D modeling of battery pack is performed and numerical results are validated by test results under two extreme but very typical operation conditions for real applications. The first one, to be specific, is the heating condition that heats battery pack from an initial temperature of $-20 \text{ }^\circ\text{C}$ to a minimum temperature of $5 \text{ }^\circ\text{C}$ in the winter, and the second one is the cooling condition that normally is a case when EVs are driving uphill of 3% slope with a constant speed of 120 km/h for 35 minutes in the summer. The objective for the heating condition is to obtain the maximum temperature difference ΔT_{max} and the heating time Δt , whereas the objective for the cooling condition is to obtain the maximum temperature difference ΔT_{max} and the maximum temperature T_{max} of battery cells. These objectives are important criteria of EV battery packs and thus need to meet some requirements for real engineering problems. Note that these requirements may differ slightly among different automobile manufactures.

After the 3D battery pack model is validated, it is then used to analyze the importance of a number of parameters, and, importantly, to determine the optimal parameters for battery packs in heating and cooling conditions under some practical engineering requirements.

4.1. Model description of the EV battery pack

The 3D model of the EV battery pack with cooling system underneath is shown in Fig. 6, where the thermal conductive pad is inserted in-between them. The inlet and outlet are sketched in the figure as well, and the outlet is with pressure of 1 atm. The computational model consists of 21 million elements. Thermal physical properties of lithium-ion battery are shown in Table 2. Heat conduction between neighboring cells, and between cells and thermal conductive pad is taken into account. Moreover, convective heat between cells and environment, and between coolant and thermal conductive pad is also under consideration. The heat transfer

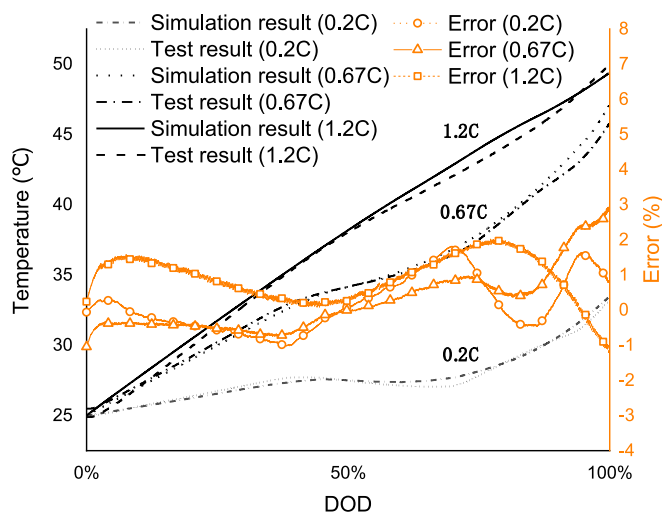


Fig. 5. Comparison between simulation and test results at different discharging rates.

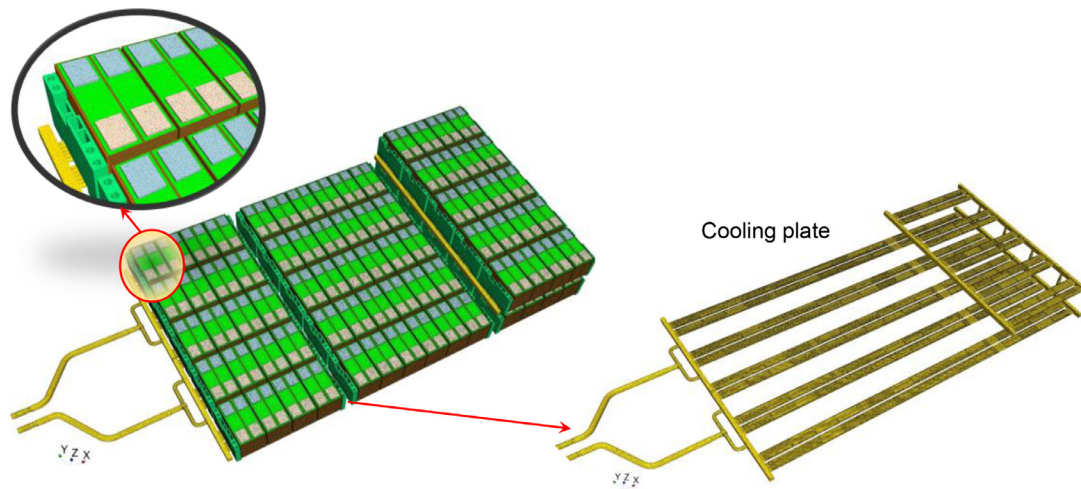


Fig. 6. 3D model of battery pack with cooling system underneath.

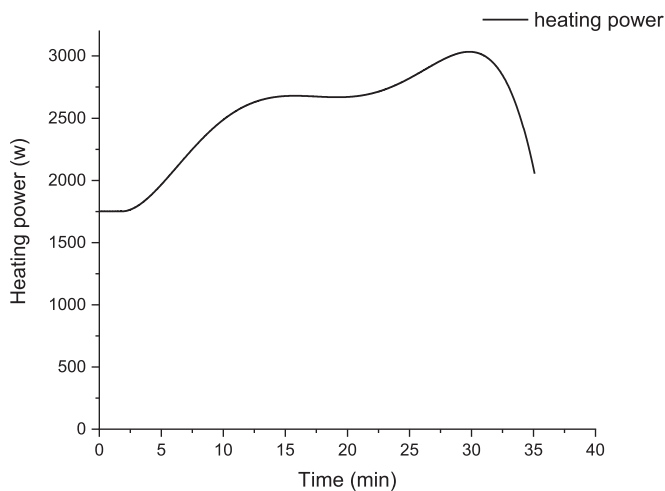


Fig. 7. The heating power of the EV battery pack in cooling condition.

via radiation is relatively small in battery thermal modeling and is thus neglected.

In heating condition, the heat source is mainly from the high-temperature coolant, which is heated by a high voltage heater (HVH) and pumped into the cooling system with an initial temperature that is much higher than cells' temperature. It needs to mention that the heat generated from battery cells is not comparable with the one from the coolant because the discharging rate is small in this operation condition. While in cooling condition, the heat source is the battery cells whose initial temperature is much higher than the coolant and is working with a high discharging rate. In both operation conditions, the heat generation model as stated in Section 3.1 is applied to describe the amount of heat generated, and note that the obtained heating power from the EV battery pack in cooling condition is shown in Fig. 7.

For real engineering problems, the specific parameters for boundary and initial conditions such as inlet flow, inlet temperature and so on for the heating and cooling conditions are shown in Table 3.

4.2. Numerical validation of the battery pack

4.2.1. Validation in heating condition

The summary of simulation and test results are compared in Table 4. It can be seen that when the minimum temperature arrives at 5 °C, the maximum temperature differences ΔT_{max} for simulation and test results are 5.7 °C and 7.0 °C, respectively. Their absolute error is 1.3 °C and they are in good agreement for engineering problems. It should be mentioned that the small difference is due to the simplification of 3D thermal model and the temperature is acquired at the center of cells' side face (depicted in Fig. 4). The heating time Δt from numerical result that of 45.4 min also agrees well with the test result that of 44.3 min, and the relative error for simulation is 2.48%. The temperature histories for both simulation and test results are shown in Fig. 8, where simulation results for both the maximum and minimum match test results in the entire heating period. The errors that defined by the current gap (between the numerical and test results) to the total temperature increment during the course are also plotted in Fig. 8, where it is clear that the errors for the numerical results of both T_{max} and T_{min} are small. The reason to select the total temperature increment for error calculation is that only the final state of temperature is used for validation, and because of the same initial temperature, the total temperature increment is thus important. It needs to mention that the initial temperature of the test results is moved from -19 °C to -20 °C for a better comparison, and the test results that are in zigzag is because of the sensors resolution embedded in mass production EVs limited to 1 °C.

The temperature distribution of the battery pack in heating condition is shown in Fig. 9, where cells located near the inlet are in higher temperature than those close to the outlet and in the

Table 2
Thermal physical properties of lithium-ion battery.

Material	Density (kg/ m ³)	Thermal conductivity (W/m/K)	Specific heat (J/kg/K)	Dynamic viscosity (Pa/s)
Al	2710.0	155.0	890.0	/
Cu	8940.0	398.0	386.0	/
Thermal conductive pad	2400.0	2	2200.0	/
Coolant	1069.4	0.4	3358.0	0.0034

Table 3

Basic parameters for two operation conditions.

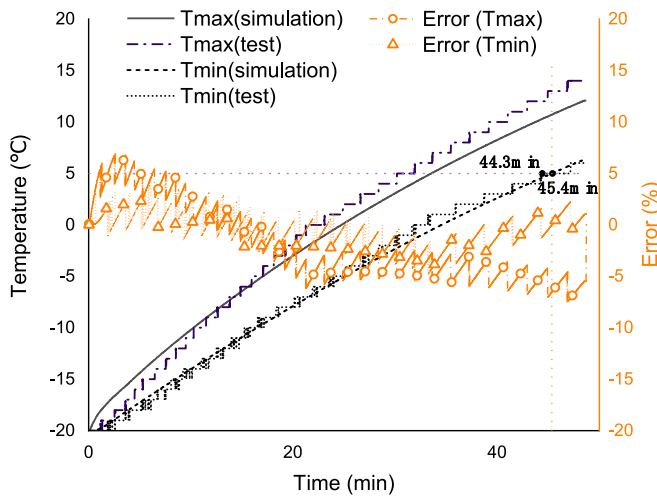
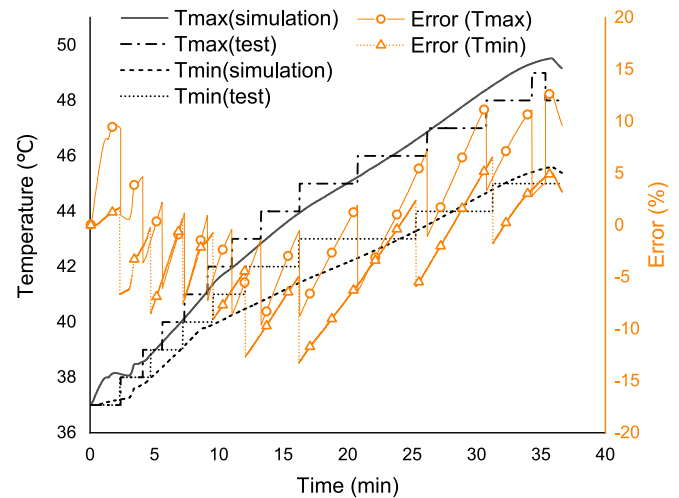
	Inlet temp. (°C)	Inlet flow (L/min)	Initial temp. of battery (°C)	Environmental temp. (°C)
Heating condition	40	10	-20	-20
Cooling condition	25	10	37	40

Table 4

Summary of simulation and test results in heating condition.

	Initial temperature (°C)			Final temperature (°C)			Heating time Δt (min)
	T_{max}	T_{min}	ΔT_{max}	T_{max}	T_{min}	ΔT_{max}	
Simulation	-20.0	-20.0	0	10.7	5.0	5.7	45.4
Test	-19.0	-20.0	1.0	13.0	5.0	7.0(8.0)	44.3

* Notes: 7.0 is the real temperature difference considering the initial temperature gap is 1.

**Fig. 8.** Comparison between numerical and test results in heating condition.**Fig. 10.** Comparison between numerical and test results in cooling condition.

upper layer. This is because the inlet flow comes in with an initial higher temperature of 40 °C (shown in Table 3) but its temperature would drop as the heat transfers from the coolant to cells whose initial temperature is -20 °C.

4.2.2. Validation in cooling condition

The summary of simulation and test results are compared in Table 5. It can be seen that when the battery pack is working in cooling condition for 35 minutes, the maximum temperature differences ΔT_{max} for simulation and test results are 3.9 °C and 4.0 °C, respectively. Their absolute error is 0.1 °C and they are in very

good agreement for engineering problems. The maximum temperature rise from numerical result that of 12.5 °C also agrees very well with the test result that of 12.0 °C, and the relative error for numerical result is 4.17%. The temperature histories for both simulation and test results are shown in Fig. 10, where simulation results for both the maximum and minimum match test results in the entire cooling period. The errors that defined by the current gap (between the numerical and test results) to the total temperature increment during the course are also plotted in Fig. 10, where it is clear that the errors for the numerical results of both T_{max} and

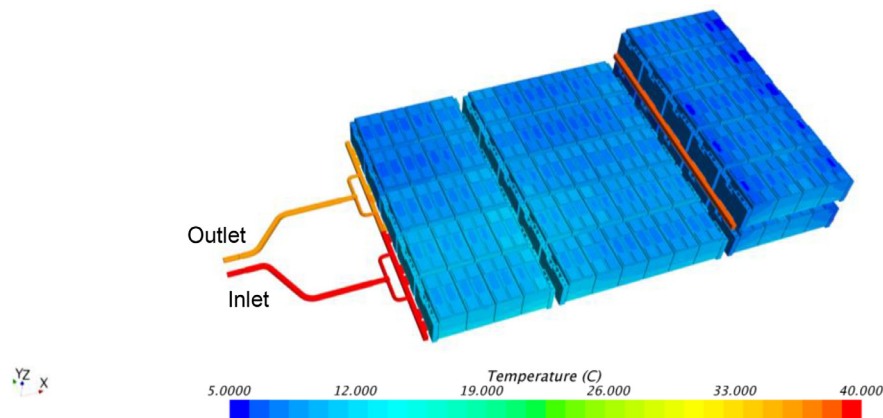
**Fig. 9.** Temperature distribution of battery pack in heating condition.

Table 5
Summary of simulation and test results in cooling condition.

	Initial temperature (°C)			Final temperature (°C)			Max Temp. rise (°C)
	T_{max}	T_{min}	ΔT_{max}	T_{max}	T_{min}	ΔT_{max}	
Simulation	37.0	37.0	0.0	49.5	45.6	3.9	12.5
Test	37.0	33.0	4.0	49.0	41.0	4.0 [*] (8.0)	12.0

* Notes: 4.0 is the real temperature difference considering the initial temperature gap is 4.0.

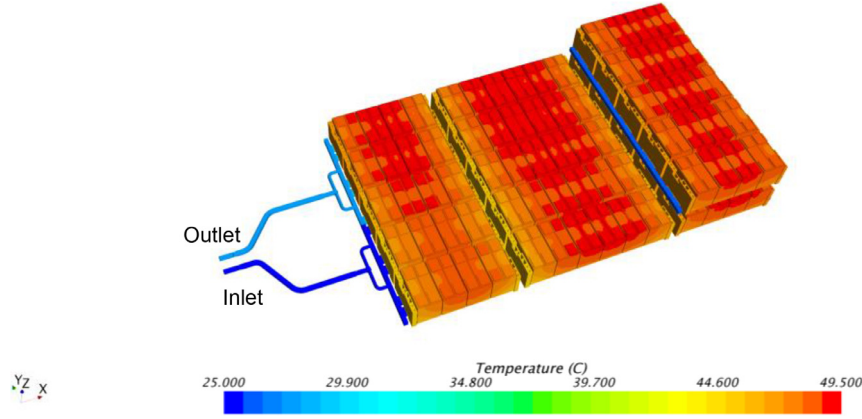


Fig. 11. Temperature distribution of battery pack in cooling condition.

T_{min} are relatively small in the entire working condition. It needs to mention that the initial temperature of the test results is moved from 33.0 °C to 37.0 °C for a better comparison.

The temperature distribution of the battery pack in cooling condition is shown in Fig. 11, where cells located near the inlet are in lower temperature than those close to the outlet and in the upper layer. This is because the inlet flow comes in with an initial lower temperature of 25 °C (shown in Table 4.2) but its temperature would rise as the heat transfers from the battery cells to coolant. Note that the initial temperature of battery pack is 37 °C as shown in Table 4.2.

4.3. Sensitive analysis

Since the maximum temperature difference ΔT_{max} , the heating time Δt and the maximum temperature T_{max} are affected by inlet flow, inlet temperature, conductive coefficients of the thermal conductive pad (k_p) and the cell in X (k_x), Y (k_y) and Z (k_z) directions, these parameters are studied to investigate their sensitivity to EV battery pack's performance. The above-mentioned 3D thermal model is set as the base model, and all these six parameters are varying from their reference values by 20% and 40% in both positive and negative directions. The variant models are conducted out in both heating and cooling conditions to evaluate battery pack's performance.

4.3.1. In heating condition

In heating condition, the impact of changing variables on the maximum temperature difference ΔT_{max} is shown in Fig. 12. From this figure, it can be seen that ΔT_{max} goes up with the increase of inlet temperature and k_z , while it drops with the increase of the remaining four variables. It is also apparent that ΔT_{max} is much more sensitive to the change of inlet temperature than the inlet flow, which is more sensitive than the other variables.

The impact of changing variables on the heating time Δt is shown in Fig. 13. From the figure, the heating time Δt goes down with the increase of all six variables. Among them, inlet temperature is notably much more sensitive to Δt than inlet flow, and both are more significant than the others.

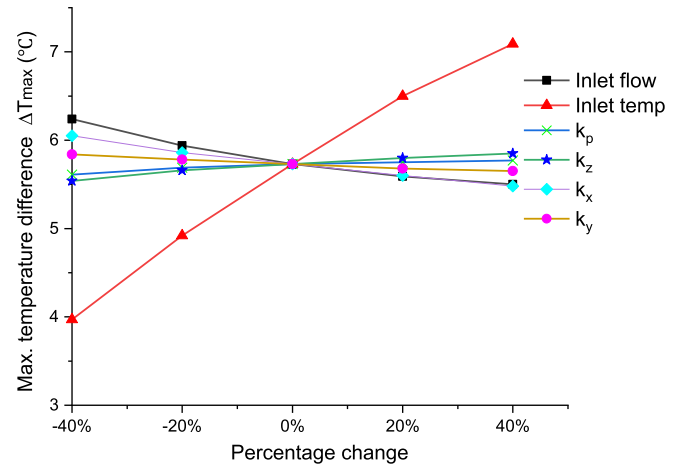


Fig. 12. Impact of changing variables on the max temperature difference ΔT_{max} in heating condition.

4.3.2. In cooling condition

In cooling condition, the impact of changing variables on the maximum temperature T_{max} is shown in Fig. 14. From this figure, it can be seen that T_{max} goes up with the increase of inlet temperature, while it drops with the increase of the remaining five variables. It is also apparent that T_{max} is much more sensitive to the change of inlet temperature than inlet flow, which is more sensitive than the other variables.

The impact of changing variables on the maximum temperature difference ΔT_{max} is shown in Fig. 15. From the figure, ΔT_{max} goes down with the increase of all six variables. Among them, inlet temperature is much more sensitive to ΔT_{max} than the others.

4.4. Optimization analysis

The maximum temperature T_{max} and the maximum temperature difference ΔT_{max} are crucial to the EV battery's safety, performance and life, and thus they have to be kept in a certain range

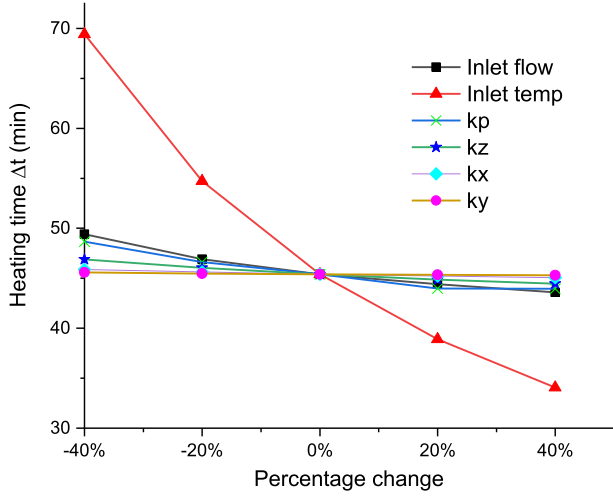


Fig. 13. Impact of changing variables on the heating time Δt in heating condition.

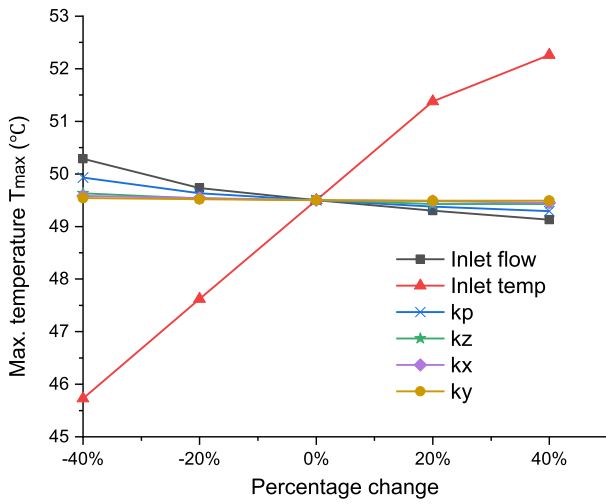


Fig. 14. Impact of changing variables on the max temperature T_{max} in cooling condition.

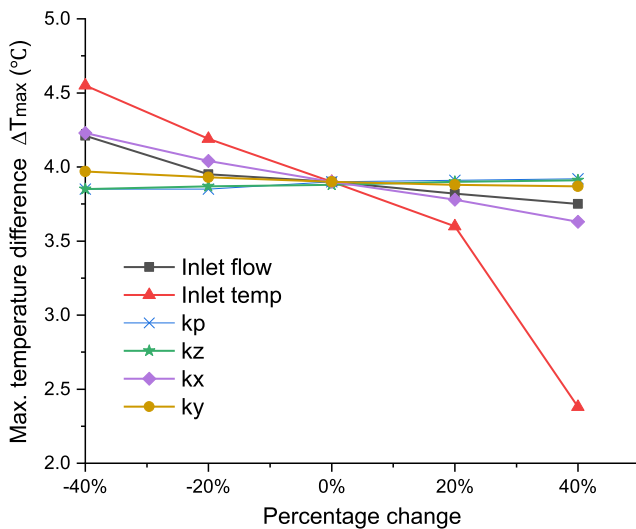


Fig. 15. Impact of changing variables on the temperature difference ΔT_{max} in cool- ing condition.

and meet the real engineering requirements. The sensitive analysis in Section 4.3 concludes that the inlet temperature and inlet flow are the top two significant parameters for the maximum temperature T_{max} , the maximum temperature difference ΔT_{max} and heating time Δt in both heating and cooling conditions. Higher inlet temperature needs a HVH consuming more power in heating condition, and lower inlet temperature requires more power on the chiller in cooling condition. For both conditions, larger inlet flow means more power consumed on the pump. To keep the EV battery pack working with the minimum energy consumed for long driving distance, an optimization analysis for inlet temperature and inlet flow is very crucial. Essentially, the objective for the optimization is to fulfill the following requirements

$$\begin{cases} \Delta t \leq 50 \text{ min} \ \& \ \Delta T_{max} \leq 6^\circ\text{C} \text{ for heating condition} \\ T_{max} \leq 47^\circ\text{C} \ \& \ \Delta T_{max} \leq 4^\circ\text{C} \text{ for cooling condition} \end{cases} \quad (13)$$

The optimization proceeds by firstly finding out the suitable inlet temperature range for each inlet flow, and then seek the common range among those obtained ranges. As the inlet temperature range narrows down, the inlet flow can be determined.

As stated in Section 4.3 that the inlet temperature and inlet flow vary from the base model by 40% in both positive and negative directions, it means the inlet temperature for heating and cooling conditions range from 24 to 56 °C and from 15 to 35 °C, separately, and the inlet flow ranges from 6 to 14 L/min for both operation conditions.

4.4.1. Optimization in heating condition

The heating time Δt and the maximum temperature difference ΔT_{max} obtained from the 3D battery pack model are as a result of the selection of inlet temperature and inlet flow, which is plotted in Fig. 16. The optimization starts from checking the suitable inlet range for the inlet flow of 14 L/min. To meet the requirement of $\Delta t \leq 50$ minutes as stated in Eq. (13), the inlet temperature has to be greater than 36.5 °C as shown by A_1 in Fig. 16. On the other hand, the inlet temperature has to be less than 45.5 °C as shown by A_2 in Fig. 16 under the constraint of $\Delta T_{max} \leq 6^\circ\text{C}$ as stated in Eq. (13) for heating condition. Therefore, the inlet temperature has to range from 36.5 °C to 45.5 °C for the inlet flow of 14 L/min in heating condition. Similarly, for the inlet flow of 10 L/min, the inlet temperature has to be greater than 38.6 °C (shown in B_1) to meet the heating time requirement ($\Delta t \leq 50$ minutes), and less than 43.0 °C (shown in B_2) to meet the maximum temperature difference requirement ($\Delta T_{max} \leq 6^\circ\text{C}$). The inlet temperature range for the inlet flow of 10 L/min is from 38.6 °C to 43.0 °C. A similar analysis to the inlet flow of 8 L/min finds out that C_1 and C_2 are almost in a line with the inlet temperature being of 40.0 °C, and to the inlet flow of 6 L/min, no feasible inlet temperature range can be found to satisfy the constraints as stated in Eq. (13) in heating condition. The optimal selection for the inlet temperature and flow is to seek the common range among the above-mentioned ranges, and it can be found that when inlet temperature is 40.0 °C and inlet flow is 8 L/min, the battery pack can achieve the best performance in terms of the minimum energy consumption for thermal management system.

4.4.2. Optimization in cooling condition

The maximum temperature T_{max} and the maximum temperature difference ΔT_{max} obtained from the 3D battery pack model are plotted in Fig. 17 as a result of the selection of inlet temperature and flow in cooling condition. The optimization starts from inlet flow of 14 L/min as well. To the requirement of $T_{max} \leq 47^\circ\text{C}$ as stated in Eq. (13), the inlet temperature has to be less than 25.3 °C as shown in A_1 in Fig. 17. It is found from this figure that, even

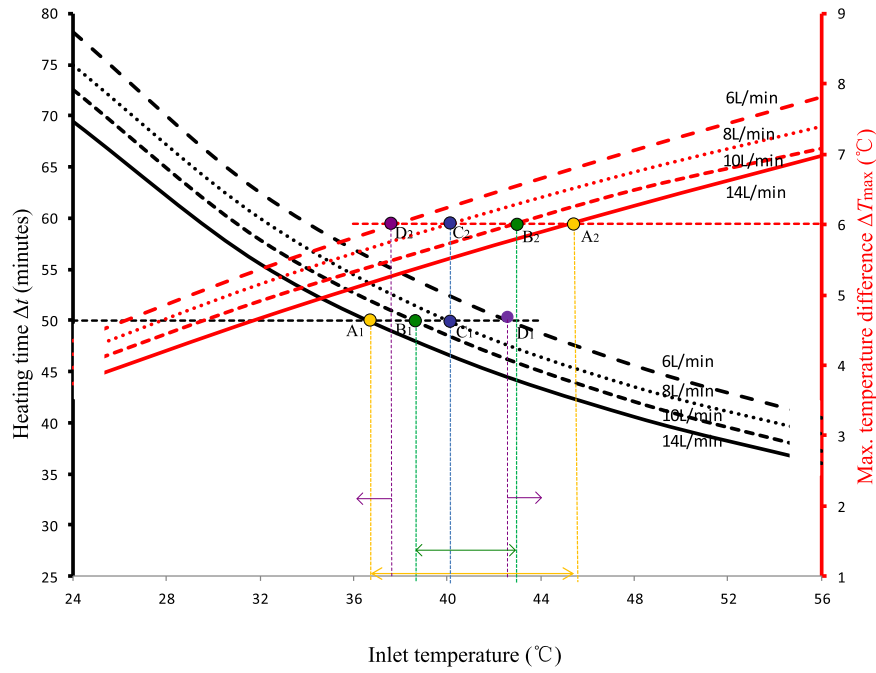


Fig. 16. Optimal selection for inlet temperature and inlet flow in heating condition.

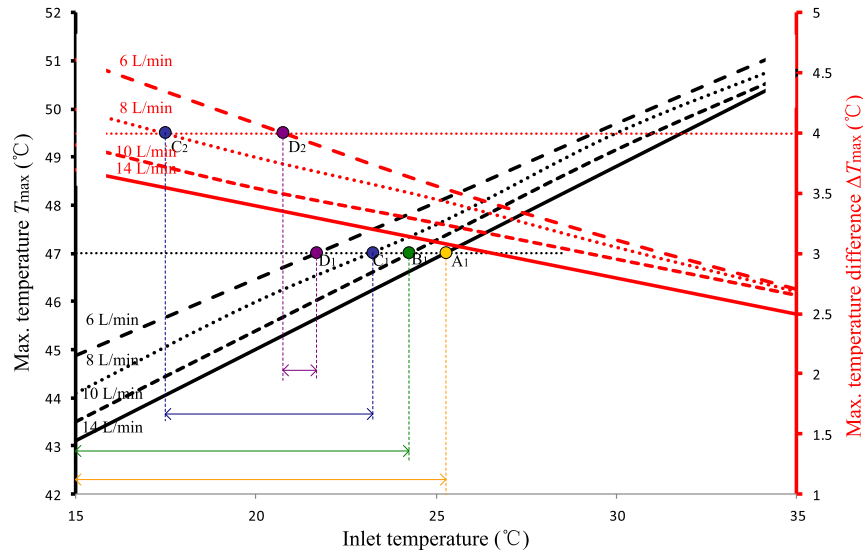


Fig. 17. Optimal selection for inlet temperature and inlet flow in cooling condition.

at the lowest temperature of 15°C, the maximum temperature difference ΔT is less than the requirement of $\Delta T_{max} \leq 4^\circ\text{C}$ as described by Eq. (13). Similar analysis to the inlet flow of 10 L/min finds that the inlet temperature has to be less than 24.2 °C (shown in B_1) to meet the maximum temperature requirement, and in the range from 15 °C to 24.2 °C, it always fulfills the constraint of $\Delta T_{max} \leq 4^\circ\text{C}$. When the inlet flow are 8 L/min and 6 L/min, the corresponding feasible inlet temperature ranges are from 17.5 °C to 23.0 °C and from 20.9 °C to 22.0 °C, separately, under the constraints stated by Eq. (13). The common range from those obtained ranges is from 20.9 °C to 22.0 °C and the best selection for inlet flow and temperature are 6 L/min and 22.0 °C, separately, in terms of the least energy consumption by the thermal management system.

5. Conclusions

Thermal analysis for an EV battery pack is conducted at two extreme operation conditions for real engineering problems and optimization on two important factors is performed. The Bernardi's heat generation model is employed and the reversible heat is taken into account. The reversible heat is related to the entropy coefficient and it is tested and presented as a relation to SOC. The specific heat capacity and the relationship between the heat and the temperature rise of battery cells are tested by the EV-ARC for thermal analysis of battery packs. The heat generation model is validated by tests at 0.2 C, 0.67 C and 1.2 C discharging rates, and it is found that the theoretical results with the reversible heat taken into account match better with the test results than the one with-

out the consideration of reversible heat. In addition, results from this comparison also reveal that the irreversible heat is sensitive to large discharging rate whereas the reversible heat plays an important role in the case of small discharging rate. Numerical results from a battery cell also validate the heat generation model at three different discharging rates and confirm the importance of the reversible heat. Then the validated cell model is applied to a real EV battery pack with cooling system underneath to study its thermal behavior at heating and cooling conditions. Numerical results from these two operation conditions are in good agreement with the test results and it confirms the accuracy of the thermal battery pack model. Several key factors important to the thermal behavior of the EV battery pack are studied by performing sensitivity analysis at two extreme operation conditions. It can be found that the inlet temperature and the inlet flow are the top two significant factors and the other four are comparatively negligible. For better thermal performance of the EV battery pack, optimization analysis at the two realistic operation conditions is conducted to determine the optimal parameters for the inlet temperature and the inlet flow. This optimization process is straightforward to implement and provides a means to find out the best input parameters for real engineering applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Z. Shang, H. Qi, X. Liu, et al., Structural optimization of lithium-ion battery for improving thermal performance based on a liquid cooling system, *Int. J. Heat Mass Transf.* 130 (2019) 33–41.
- [2] Z. Rao, S. Wang, A review of power battery thermal energy management, *Renew. Sustain. Energy Rev.* 15 (2011) 4554–4571.
- [3] C. Qi, Y. Zhu, F. Gao, et al., Mathematical model for thermal behavior of lithium ion battery pack under overcharge, *Int. J. Heat Mass Transf.* 124 (2018) 552–563.
- [4] X. Feng, L. Lu, M. Ouyang, et al., A 3D thermal runaway propagation model for a large format lithium ion battery module, *Energy* 115 (2016) 194–208.
- [5] D. Bernardi, E. Pawlikowski, J. Newman, A general energy balance for battery systems, *J. Electrochem. Soc.* 132 (1985) 5–12.
- [6] S. Al Hallaj, H. Maleki, J.S. Hong, et al., Thermal modeling and design considerations of lithium-ion batteries, *J. Power Sources* 83 (1999) 1–8.
- [7] J. Zhang, J. Huang, Z. Li, et al., Comparison and validation of methods for estimating heat generation rate of large-format lithium-ion batteries, *J. Therm. Anal. Calorimetry* 117 (2014) 447–461.
- [8] K. Yang, D.H. Li, S. Chen, et al., Thermal behavior of nickel/metal hydride battery during charging and discharging, *J. Therm. Anal. Calorimetry* 95 (2009) 455–459.
- [9] Y. Inui, Y. Kobayashi, Y. Watanabe, et al., Simulation of temperature distribution in cylindrical and prismatic lithium ion secondary batteries, *Energy Conv. Manag.* 48 (2007) 2103–2109.
- [10] J. Smith, M. Hinterberger, P. Hable, et al., Simulative method for determining the optimal operating conditions for a cooling plate for lithium-ion battery cell modules, *J. Power Sources* 267 (2014) 784–792.
- [11] S.C. Chen, C.C. Wan, Y.Y. Wang, Thermal analysis of lithium-ion batteries, *J. Power Sources* 140 (2005) 111–124.
- [12] A. Jarrett, I.Y. Kim, Design optimization of electric vehicle battery cooling plates for thermal performance, *J. Power Sources* 196 (2011) 10359–10368.
- [13] A. Jarrett, I.Y. Kim, Influence of operating conditions on the optimum design of electric vehicle battery cooling plates, *J. Power Sources* 245 (2014) 644–655.
- [14] K. Chen, M. Song, W. Wei, et al., Design of the structure of battery pack in parallel air-cooled battery thermal management system for cooling efficiency improvement, *Int. J. Heat Mass Transf.* 132 (2019) 309–321.
- [15] K. Chen, S. Wang, M. Song, et al., Structure optimization of parallel air-cooled battery thermal management system, *Int. J. Heat Mass Transf.* 111 (2017) 943–952.
- [16] G.G. Botte, V.R. Subramanian, R.E. White, Mathematical modeling of secondary lithium batteries, *Electrochimica Acta* 45 (2000) 2595–2609.
- [17] Z. Li, J. Huang, B.Y. Liaw, et al., On state-of-charge determination for lithium-ion batteries, *J. Power Sources* 348 (2017) 281–301.
- [18] H.-G. Schweiger, O. Obeidi, O. Komesker, et al. Comparison of several methods for determining the internal resistance of lithium ion cells, *Sensors (Basel, Switzerland)* 10 (2010) 5604–5625.
- [19] K.E. Thomas, C. Bogatu, J. Newman, Measurement of the entropy of reaction as a function of state of charge in doped and undoped lithium manganese oxide, *J. Electrochem. Soc.* 148 (2001) A570–A575.
- [20] Y. Zheng, M. Ouyang, L. Lu, et al. Understanding aging mechanisms in lithium-ion battery packs: from cell capacity loss to pack capacity evolution, *J. Power Sources* 278 (2015) 287–295.
- [21] M. Petzl, M.A. Danzer, Advancements in OCV measurement and analysis for lithium-ion batteries, *IEEE Trans. Energy Conv.* 28 (2013) 675–681.
- [22] W. Dreyer, J. Jamnik, C. Gohlke, et al. The thermodynamic origin of hysteresis in insertion batteries, *Nat. Mater.* 9 (2010) 448–453.
- [23] M.A. Roscher, D.U. Sauer, Dynamic electric behavior and open-circuit-voltage modeling of LiFePO₄-based lithium ion secondary batteries, *J. Power Sources* 196 (2011) 331–336.